

Fourier Transform & Stokes' Theorem

Lecture 08 · Detailed study notes with step-by-step proofs, worked solutions and two case studies

About this document. Everything here is built from the handwritten lecture notes and the two case-study handouts (noise removal with the Fourier Transform, and MRI field circulation with Stokes' Theorem). The derivations follow the lecturer's steps, but each step is explained in full. Where the notes contained a slip or left an exercise open, this is flagged in a **highlighted box**.

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Part 1 · Fourier Transform

1.1 Definition

Let $f(x)$ be an **integrable function** over the interval $(-\infty, \infty)$, meaning the area under $|f(x)|$ is finite. Its Fourier Transform is the function $F(k)$ defined by

$$\mathcal{F}\{f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx} f(x) dx = F(k)$$

The transform converts a function of position x (or time) into a function of the wave-number k (or frequency). It tells us *how much* of each oscillation e^{ikx} is hidden inside $f(x)$.

Convention used in these notes: kernel e^{+ikx} and normalising factor $1/\sqrt{2\pi}$. Other books use e^{-ikx} ; the property formulas below are specific to the convention above.

1.2 Shifting and scaling properties

Three rules let us find the transform of a modified function without recomputing the integral each time:

#	Property	Formula	In words
1	Shift in x	$\mathcal{F}\{f(x-a)\} = e^{ika} F(k), a > 0$	Moving the graph right by a only multiplies the transform by the phase factor e^{ika} .
2	Shift in k (modulation)	$\mathcal{F}\{e^{iax} f(x)\} = F(k+a), a > 0$	Multiplying $f(x)$ by e^{iax} slides the whole transform along the k -axis.
3	Scaling	$\mathcal{F}\{f(cx)\} = \frac{1}{c} F\left(\frac{k}{c}\right)$	Squeezing f in x ($c > 1$) stretches F in k , and vice-versa. Taken for $c > 0$.

Note on the lecture heading. The list of properties writes the second one as $e^{iak} f(x)$, but the proof (and the correct statement) uses $e^{iax} f(x)$, with the exponent multiplying x . For $c < 0$ in property 3 the factor $1/c$ becomes $1/|c|$.

1.3 Proof of Property 1: $\mathcal{F}\{f(x-a)\} = e^{ika} F(k)$

Step 1. Write the definition with $f(x-a)$ inside the integral.

$$\mathcal{F}\{f(x-a)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx} f(x-a) dx$$

Step 2. Substitute $x' = x - a$. Then $x = x' + a$ and $dx = dx'$ (a is a constant). Because a is finite, the limits do not change: $x \rightarrow -\infty$ gives $x' \rightarrow -\infty$, and $x \rightarrow +\infty$ gives $x' \rightarrow +\infty$.

$$\begin{aligned} x' = x - a &\Rightarrow x = x' + a, & dx &= dx' \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ik(x'+a)} f(x') dx' \end{aligned}$$

Step 3. Expand the exponent and split it into a product.

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx' + ika} f(x') dx' = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx'} e^{ika} f(x') dx'$$

Step 4. e^{ika} does not depend on x' , so it comes outside the integral. What is left inside the brackets is exactly the definition of $F(k)$.

$$= e^{ika} \left\{ \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx'} f(x') dx' \right\} = e^{ika} F(k)$$

Hence proved: $F\{f(x - a)\} = e^{ika} F(k)$.

1.4 Proof of Property 2: $F\{e^{iax} f(x)\} = F(k + a)$

Step 1. Insert the extra factor into the definition.

$$\mathcal{F}\{e^{iax} f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx} e^{iax} f(x) dx$$

Step 2. Combine the two exponentials: $e^{ikx} \cdot e^{iax} = e^{i(k+a)x}$.

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i(k+a)x} f(x) dx$$

Step 3. This is the original definition with k replaced by $(k + a)$, so it equals $F(k + a)$.

$$= F(k + a)$$

Proved: $F\{e^{iax} f(x)\} = F(k + a)$.

1.5 Proof of Property 3: $F\{f(cx)\} = (1/c) F(k/c)$

Step 1. Start from the definition.

$$\mathcal{F}\{f(cx)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx} f(cx) dx$$

Step 2. Substitute $x' = cx$ ($c > 0$). Then $x = x'/c$ and $dx = dx'/c$. The limits stay $-\infty$ and $+\infty$. Also $e^{ikx} = e^{i(k/c)x'}$.

$$\begin{aligned} x' = cx &\Rightarrow x = \frac{x'}{c}, & dx &= \frac{dx'}{c} \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i(k/c)x'} f(x') \frac{dx'}{c} \end{aligned}$$

Step 3. Take the constant $1/c$ outside; the remainder is F evaluated at k/c .

$$= \frac{1}{c} \left\{ \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i(k/c)x'} f(x') dx' \right\} = \frac{1}{c} F\left(\frac{k}{c}\right)$$

Proved: $F\{f(cx)\} = (1/c) F(k/c)$.

1.6 Worked examples from the lecture

Example A: $F\{e^{i2x} f(x)\}$ (Property 2 with $a = 2$)

$$\mathcal{F}\{e^{i2x} f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx} e^{i2x} f(x) dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i(k+2)x} f(x) dx = F(k + 2)$$

Example B: $\mathcal{F}\{f(x - 20)\}$ (Property 1 with $a = 20$)

$$\begin{aligned}
 x' = x - 20 &\Rightarrow x = x' + 20, & dx &= dx' \\
 \mathcal{F}\{f(x - 20)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ik(x' + 20)} f(x') dx' \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx'} e^{i20k} f(x') dx' \\
 &= e^{i20k} \left\{ \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx'} f(x') dx' \right\} = e^{i20k} F(k)
 \end{aligned}$$

Example C: $\mathcal{F}\{f(5x)\}$ (Property 3 with $c = 5$)

$$\begin{aligned}
 x' = 5x &\Rightarrow x = \frac{x'}{5}, & dx &= \frac{dx'}{5}; & x \rightarrow \pm \infty &\Rightarrow x' \rightarrow \pm \infty \\
 \mathcal{F}\{f(5x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i(k/5)x'} f(x') \frac{dx'}{5} \\
 &= \frac{1}{5} \left\{ \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i(k/5)x'} f(x') dx' \right\} = \frac{1}{5} F\left(\frac{k}{5}\right)
 \end{aligned}$$

1.7 Practice exercises from the lecture, with answers

The notes end the properties section with three exercises. Solutions:

(1) $\mathcal{F}\{f(x - 100)\}$

Property 1 with $a = 100$:

$$\mathcal{F}\{f(x - 100)\} = e^{i100k} F(k)$$

(2) $\mathcal{F}\{e^{i10x} f(x)\}$

Property 2 with $a = 10$:

$$\mathcal{F}\{e^{i10x} f(x)\} = F(k + 10)$$

(3) $\mathcal{F}\{f(100 - x)\}$

This one is a reflection plus a shift (x is replaced by $-x$), so Property 1 alone does not cover it. Substitute $u = 100 - x$ directly:

$$\begin{aligned}
 u = 100 - x &\Rightarrow x = 100 - u, & dx &= -du; & x \rightarrow -\infty &\Rightarrow u \rightarrow +\infty, & x \rightarrow +\infty &\Rightarrow u \rightarrow -\infty \\
 \mathcal{F}\{f(100 - x)\} &= \frac{1}{\sqrt{2\pi}} \int_{+\infty}^{-\infty} e^{ik(100 - u)} f(u) (-du) \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ik(100 - u)} f(u) du
 \end{aligned}$$

The minus sign from $dx = -du$ cancels against the swapping of the limits. Factor out the constant e^{i100k} :

$$= e^{100ik} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iku} f(u) du = e^{100ik} F(-k)$$

Answers: (1) $e^{i100k} F(k)$ · (2) $F(k + 10)$ · (3) $e^{i100k} F(-k)$.

1.8 Fourier transform of piece-wise functions (important)

When $f(x)$ is defined by different formulas on different intervals, split the integral at the break-points. If $f_1(x)$ applies on $(-\infty, 0]$ and $f_2(x)$ applies on $(0, \infty)$:

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^0 f_1(x) dx + \int_0^{\infty} f_2(x) dx$$

Example 1

$$f(x) = 0, \quad -\infty < x \leq 0; \quad f(x) = e^{-x}, \quad 0 < x < +\infty$$

Step 1. Split the definition integral. On $(-\infty, 0]$ the function is 0, so that part vanishes.

$$F(k) = \frac{1}{\sqrt{2\pi}} \left[\int_{-\infty}^0 e^{ikx} (0) dx + \int_0^{\infty} e^{ikx} e^{-x} dx \right] = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{(ik-1)x} dx$$

Step 2. Integrate the exponential: the antiderivative of $e^{(ik-1)x}$ is $e^{(ik-1)x}/(ik-1)$.

$$\begin{aligned} &= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{(ik-1)x}}{ik-1} \right]_0^{\infty} \\ &= \frac{1}{\sqrt{2\pi} (ik-1)} [e^{-(1-ik)x}]_0^{\infty} \end{aligned}$$

Step 3. Evaluate the limits. The factor e^{ikx} only rotates (its size is 1), while $e^{-x} \rightarrow 0$ as $x \rightarrow \infty$, so the upper limit gives 0. The lower limit gives $e^0 = 1$.

$$= \frac{1}{\sqrt{2\pi} (ik-1)} [e^{-\infty} - e^0] = \frac{1}{\sqrt{2\pi} (ik-1)} (0 - 1)$$

Answer:

$$F(k) = -\frac{1}{\sqrt{2\pi} (ik-1)} = \frac{1}{\sqrt{2\pi} (1-ik)}$$

Example 2

$$f(x) = e^{-2x}, \quad -\infty < x \leq 0; \quad f(x) = 0, \quad 0 < x < +\infty$$

Step 1. Split the integral; this time the second piece ($x > 0$) is the zero one.

$$F(k) = \frac{1}{\sqrt{2\pi}} \left[\int_{-\infty}^0 e^{ikx} e^{-2x} dx + \int_0^{\infty} e^{ikx} (0) dx \right] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 e^{(ik-2)x} dx$$

Step 2. Integrate and substitute the limits, as written in the lecture.

$$= \frac{1}{\sqrt{2\pi} (ik-2)} [e^{(ik-2)x}]_{-\infty}^0 = \frac{1}{\sqrt{2\pi} (ik-2)} [e^0 - e^{-\infty (ik-2)}]$$

The lecture then treats the second term as 0, giving the answer $1 / (\sqrt{2\pi} (ik-2))$.

Correction needed. For $f(x) = e^{-2x}$ on $(-\infty, 0]$, the function *grows* as $x \rightarrow -\infty$ ($e^{-2x} \rightarrow \infty$). At the lower limit $|e^{(ik-2)x}| = e^{-2x} \rightarrow \infty$, so that term is **not** 0: the integral diverges and f is not integrable, so this Fourier Transform does not exist in the sense of the definition. The result above is therefore only valid for the decaying version of the problem:

$$f(x) = e^{2x} \quad (x \leq 0), \quad f(x) = 0 \quad (x > 0)$$

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 e^{(ik+2)x} dx = \frac{1}{\sqrt{2\pi}(ik+2)} [e^{(ik+2)x}]_{-\infty}^0 = \frac{1}{\sqrt{2\pi}(2+ik)}$$

Here $|e^{(ik+2)x}| = e^{2x} \rightarrow 0$ as $x \rightarrow -\infty$, so the lower limit really is 0. It is worth checking with your instructor which version was intended.

1.9 Case study: noise removal from a signal using the Fourier Transform

1. Introduction

Real-life signals such as voice or music are often corrupted by unwanted noise, which lowers quality and clarity. The Fourier Transform converts a signal from the **time domain** to the **frequency domain**, where noise is easy to identify and remove. A group of 3rd-semester BS Software Engineering students at Riphah University used this idea to propose a way of removing unwanted noise from sound, video and recordings, demonstrated here on a simple signal.

2. Given signal

The signal contains a main component and a noise component:

$$x(t) = \cos(2\pi \cdot 5t) + 0.5\cos(2\pi \cdot 20t)$$

- **5 Hz** → main signal (amplitude 1)
- **20 Hz** → noise (amplitude 0.5)

3. Objective

- Identify the frequency components of the signal
- Detect the noise frequency
- Apply the Fourier Transform to separate the components
- Remove the noise using a low-pass filter $H(f)$
- Obtain a clean signal

4. Fourier Transform analysis

The Fourier Transform of a cosine is a pair of impulses (delta functions) at plus and minus its frequency:

$$\mathcal{F}\{\cos(2\pi f_0 t)\} = \frac{1}{2}[\delta(f - f_0) + \delta(f + f_0)]$$

Applying this to each term of the given signal:

$$X(f) = \frac{1}{2}[\delta(f - 5) + \delta(f + 5)] + \frac{0.5}{2}[\delta(f - 20) + \delta(f + 20)]$$

Interpretation: peaks at ± 5 Hz are the main signal; peaks at ± 20 Hz are the noise. (This case study uses ordinary frequency f in Hz, so its transform convention differs from the k -based definition in 1.1.)

5. Filter design $H(f)$

To remove the noise a **low-pass filter** is applied:

$$H(f) = 1 \text{ for } |f| \leq 10, \quad H(f) = 0 \text{ for } |f| > 10$$

- Passes frequencies below 10 Hz (the main signal)
- Blocks frequencies above 10 Hz (the noise)

6. Apply the filter

In the frequency domain, filtering is just multiplication:

$$Y(f) = X(f) \cdot H(f) \Rightarrow Y(f) = \frac{1}{2}[\delta(f - 5) + \delta(f + 5)]$$

- Noise at 20 Hz is removed
- Only the main signal at 5 Hz remains

7. Inverse Fourier Transform

Converting back to the time domain gives the clean signal:

$$y(t) = \cos(2\pi \cdot 5t)$$

8. Results table

Frequency (Hz)	Component	After filter
5	Main signal	Kept
20	Noise	Removed

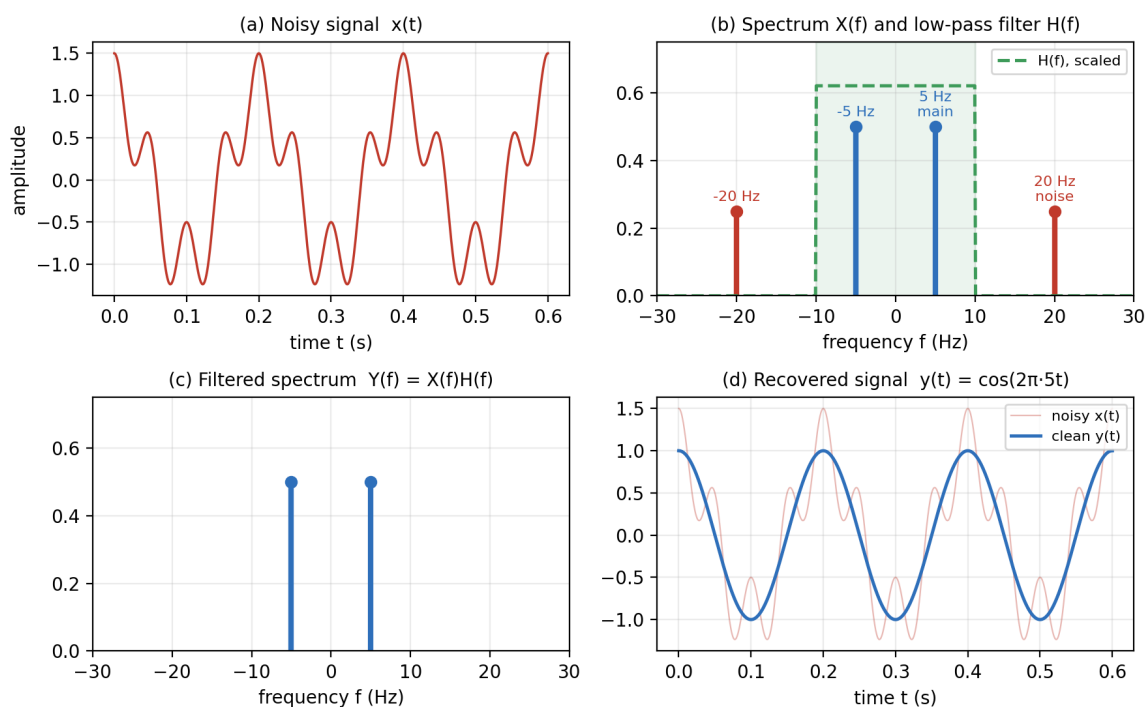


Figure 1 (added illustration, drawn from the case-study signal): (a) noisy signal, (b) its spectrum with the 10 Hz low-pass filter, (c) the filtered spectrum, (d) the recovered 5 Hz signal.

9. Conclusion

The Fourier Transform separates a signal into its frequency components. By applying a suitable filter, noise can be removed and the desired signal left intact. The method is simple, clear and effective for understanding the fundamentals of signal processing.

Part 2 · Stokes' Theorem

2.1 Statement and intuition

Stokes' Theorem: the line integral of a vector field around a closed curve is equal to the surface integral of the curl of that vector field over *any* surface bounded by the same curve.

Measuring the circulation along the boundary = Measuring the total rotation over the surface inside.

Think of the surface as a grid of tiny patches. Each patch has a small amount of rotation (curl). Along every shared interior edge, the circulation of one patch is cancelled by its neighbour travelling the opposite way, so only the circulation along the outer boundary survives. The theorem is used to **convert a surface integral into a line integral** (or the other way round), whichever is easier to compute.

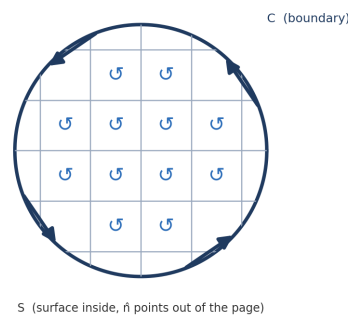


Figure 2 (added illustration): the surface S is split into small patches ($\mathcal{U}$); interior circulations cancel, leaving the circulation along C .

Mathematically:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} dS = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$$

2.2 Meaning of each symbol

Symbol	Meaning
$\vec{F}$	The vector field.
C	Closed curve (the boundary edge).
S	A surface whose boundary is the curve C .
$\nabla \times \vec{F}$	Curl of F : the local rotation of the field.
$\hat{n}$	Unit normal vector at each point of the surface (perpendicular to S).
$d\vec{r}$	Small displacement along the curve.
dS	Small area element of the surface.

Orientation: by the usual right-hand rule, C is traversed counter-clockwise when seen from the tip of $\hat{n}$. This fixes the sign of the result.

2.3 Derivation (surface lying in the xy-plane)

Left-hand side

Step 1. Write the field and the displacement in components.

$$\vec{F} = P\hat{i} + Q\hat{j} + R\hat{k}, \quad d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

Step 2. Take the dot product. Using $\hat{i}\cdot\hat{i} = \hat{j}\cdot\hat{j} = \hat{k}\cdot\hat{k} = 1$ and all mixed products $\hat{i}\cdot\hat{j} = \hat{j}\cdot\hat{k} = \hat{i}\cdot\hat{k} = 0$, only matching components survive.

$$\vec{F} \cdot d\vec{r} = (P\hat{i} + Q\hat{j} + R\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = Pdx + Qdy + Rdz$$

So the left-hand side is $\oint_C (Pdx + Qdy + Rdz)$.

Right-hand side

Step 3. Compute the curl as a determinant, then expand it.

$$\begin{aligned} \nabla \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix} \\ \nabla \times \vec{F} &= \hat{i} \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) - \hat{j} \left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z} \right) \\ &\quad + \hat{k} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \end{aligned}$$

Step 4. Assume the surface is the xy-plane. Then $z = 0$, the unit normal is $\hat{n} = \hat{k}$, and the area element is $dS = dx dy$. Dot-product rules $\hat{i}\cdot\hat{k} = 0$ and $\hat{j}\cdot\hat{k} = 0$ kill the $\hat{i}$ and $\hat{j}$ parts of the curl, and $\hat{k}\cdot\hat{k} = 1$ keeps the $\hat{k}$ part:

$$(\nabla \times \vec{F}) \cdot \hat{n} = \hat{k} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \cdot \hat{k} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

Step 5. Substitute into the surface integral.

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS = \iint_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

Step 6. Equate both sides. In the plane $z = 0$ we also have $dz = 0$ along C , so the $R dz$ term contributes nothing.

$$\begin{aligned} \oint_C (Pdx + Qdy + Rdz) &= \iint_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy \\ \oint_C \vec{F} \cdot d\vec{r} &= \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS \end{aligned}$$

Proved (for a surface in the xy-plane).

Scope of this proof. With a flat surface in the xy-plane the result is the plane case, known as **Green's Theorem**, and the last step above leans on it. For a general curved surface, Stokes' Theorem follows by cutting S into many tiny, almost-flat patches (the grid in Figure 2), applying the flat-surface result to each patch and adding up: interior edges cancel.

2.4 Case study: magnetic field circulation analysis in MRI scanners using Stokes' Theorem

1. Problem

Ms. Ayesha, a 29-year-old patient, visited a diagnostic centre for an MRI scan because of continuous headaches and dizziness. Earlier scans had shown distortions in the MRI images, raising concerns about diagnostic accuracy.

2. Medical suspicion

After reviewing the results, doctors suspected that the magnetic field inside the scanner was not uniform. Such irregularity could disturb the circulation of the magnetic field and lead to blurred or distorted images.

3. Engineer's approach

The biomedical engineer explained that magnetic fields behave like vector fields, so their circulation can be studied with vector calculus; by modelling the field inside the scanner, disturbances in its behaviour can be identified. For simplicity:

- The MRI coil is modelled as a **closed curve C**
- The surface bounded by the coil is denoted **S**
- The magnetic field is a **three-dimensional vector field**

4. Choosing a mathematical model

$$\vec{F}(x, y, z) = (-y, x, z)$$

The engineer's reasons for this choice:

- The model represents rotational magnetic behaviour
- Controlled rotation is required for proper MRI functioning
- Disturbances in rotation can lead to image distortion

5. Introducing Stokes' Theorem

Stokes' Theorem relates circulation to curl:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$$

The circulation of the field around the closed curve C equals the curl of the field integrated over the surface S enclosed by C.

Applying the theorem

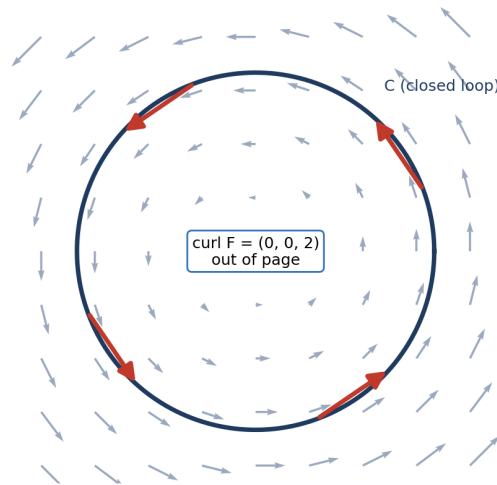
With $F = (P, Q, R) = (-y, x, z)$, the curl is computed from the determinant. Component by component:

Component	Computation	Value
$\hat{i}$	$\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} = \frac{\partial z}{\partial y} - \frac{\partial x}{\partial z}$	$0 - 0 = 0$
$\hat{j}$	$-\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) = -\left(\frac{\partial z}{\partial x} - \frac{\partial(-y)}{\partial z}\right)$	$-(0 - 0) = 0$
$\hat{k}$	$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{\partial x}{\partial x} - \frac{\partial(-y)}{\partial y}$	$1 - (-1) = 2$

$$\nabla \times \vec{F} = (0, 0, 2)$$

The engineer explained: a **non-zero curl** indicates strong magnetic field circulation, and excessive circulation suggests irregular field behaviour.

Fig (1): Magnetic field circulation around MRI coil
(grey arrows: $F = (-y, x)$, red: circulation along C)



Closed loop C represents the MRI coil; the arrows show the circulation of the field. By Stokes' Theorem this circulation is directly related to the curl over S . Irregular circulation leads to MRI image distortion.

A direct numerical check (added)

To confirm the theorem on this model, take the coil as a circle of radius r in the xy -plane, $x = r \cos t$, $y = r \sin t$, $0 \leq t \leq 2\pi$ (so $z = 0$ and $dz = 0$). **Line integral:**

$$\oint_C \vec{F} \cdot d\vec{r} = \oint_C (-y dx + x dy) = \int_0^{2\pi} (r^2 \sin^2 t + r^2 \cos^2 t) dt = 2\pi r^2$$

Surface integral (curl is $(0, 0, 2)$ and $\hat{n} = \hat{k}$, so the integrand is the constant 2; the disc has area πr^2):

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{k} dS = \iint_S 2 dA = 2 \cdot \pi r^2 = 2\pi r^2$$

Both sides agree, as Stokes' Theorem requires.

6. Comparing normal and disturbed magnetic fields

Condition	Magnetic field behaviour
Properly functioning MRI scanner	Uniform circulation, low curl
Faulty MRI scanner	Irregular circulation, high curl

7. Medical interpretation

The engineer concluded that the high curl value indicated irregular magnetic field circulation, which explained the distorted MRI images. This mathematical result supported the doctors' observations.

8. Final outcome

Doctors used this analysis to:

- Confirm magnetic field irregularities
- Schedule maintenance and recalibration of the MRI scanner
- Ensure accurate imaging for future patients

Modelling note. $F = (-y, x, z)$ is a teaching model. Its curl is the same constant $(0, 0, 2)$ everywhere, so it illustrates how curl measures rotation rather than describing a real scanner. In practice, MRI field uniformity is assessed with dedicated field-mapping and corrected by shimming and calibration.